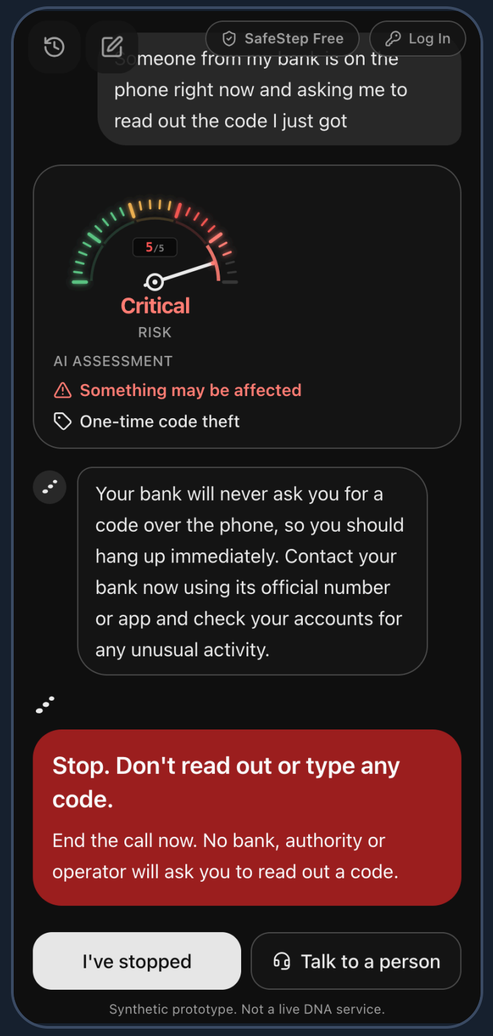
DNA CHALLENGE, SINCE AI HACKATHON 2026

SafeStep

When a call, message or link looks wrong, SafeStep tells you what to do next.

Mehrab and Arsham, Turku, 8 November 2026



THE PROBLEM

A warning says something is wrong. It doesn’t say what to do.

People panic

A caller is still on the line, a link was already clicked, money may already be gone.

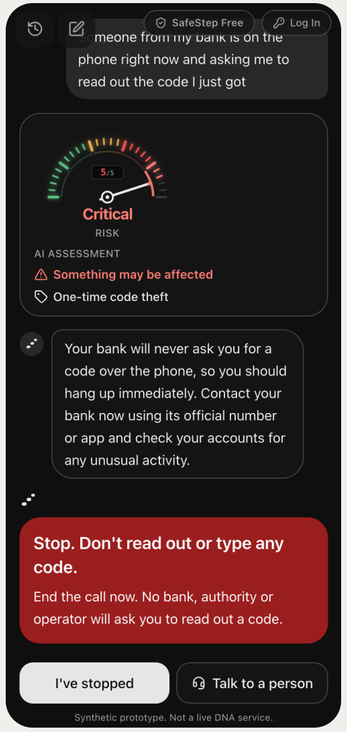
Advice is generic

The right step depends on one fact: did you share the code, did the payment leave?

Small firms are alone

Invoice fraud and account takeover hit businesses without a security team.

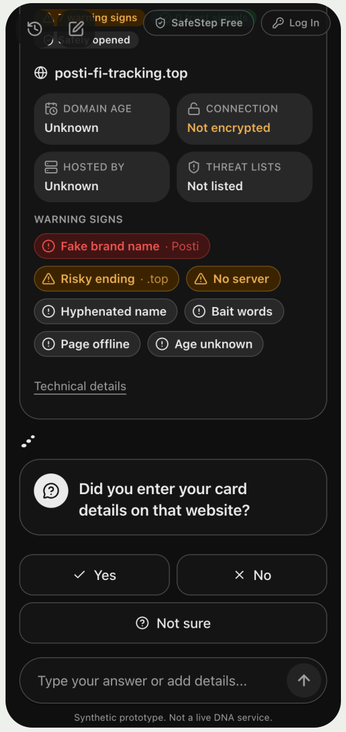
One clear next step, in three moves



1

Stop the harm

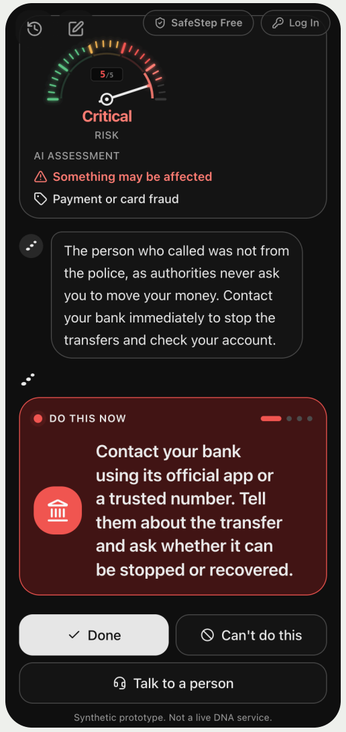
Live danger gets a Stop screen before any question.



2

Ask one question

Only if the answer changes the advice. “Not sure” is fine.



3

Guide recovery

One step at a time, and a DNA person when needed.

Finnish, English and Swedish, by text or voice. 3

HOW IT WORKS

The LLM reads. Code decides.

Mask
Codes, cards and passwords hidden first

Read
Gemini turns words into facts, each with a quote

Check
No quote in the text, no fact

Decide
58 rules pick the action and the person

Answer
Approved text only, checked before it is shown

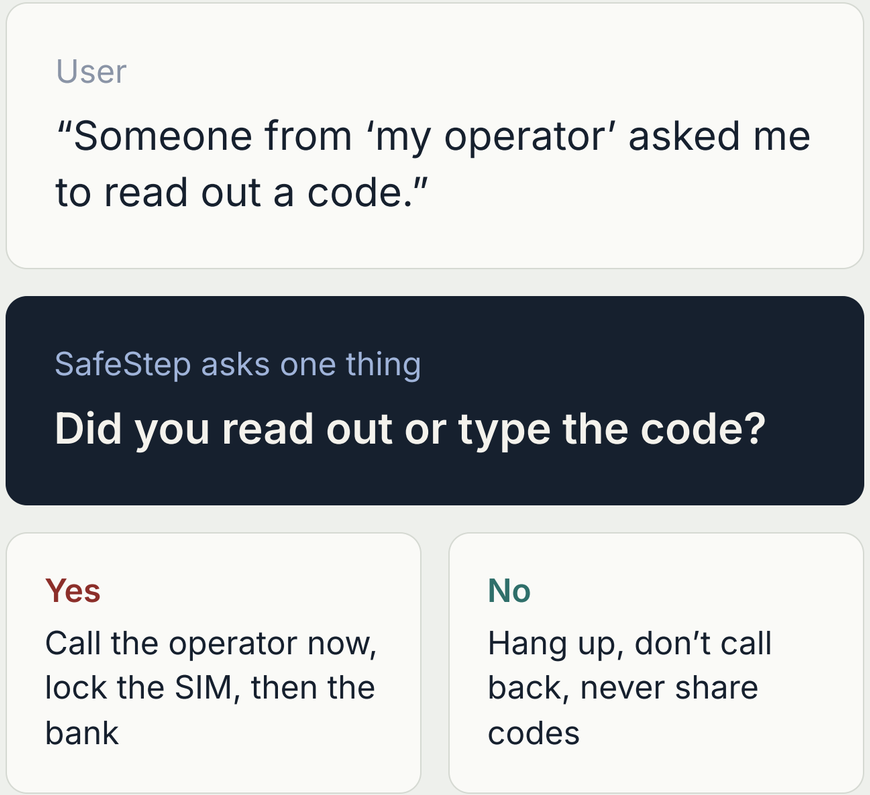
The model never chooses risk, action or who to call. If it fails, a keyword reader takes over, so advice still works with no AI at all.

DECISION FLIP

We only ask what would change the advice

For every unknown fact, the rules run twice: once with “Yes”, once with “No”. If both give the same advice, we don’t ask. If they differ, we ask that one question.

“Not sure” keeps the fact unknown. The advice then covers the worse case, written as a condition, and never claims it happened.



Asked the decisive question in 96.7% of “it depends” cases. 5

SAFETY AND PRIVACY

What SafeStep will never do

Call it safe without proof

“Nothing risky found” is never enough

Use data without consent

DNA signals only after a yes, for one session

Act on its own

No blocking or paying; every handoff is approved

Keep or ask for secrets

Codes and cards masked before the AI sees them

Obey a scam message

“Verified safe” inside a message is ignored

Open links unsafely

Plain-text fetch, no scripts, private IPs refused

0.0% unsafe statements, 0.0% consent violations on 588 cases. 6

RESULTS ON DNA'S 588 TEST CASES

Safer and calmer than the baseline

94.4%
Safe-action pass
baseline 60.5%

0%
Unsafe statements
target 0%

97.0%
Escalation recall
baseline 47.0%

9.8%
False alarms
baseline 28.0%

Scored with the organisers' evaluator v2.0.1. Weighted brief score 94.0 against a baseline of 67.1. Rules were tuned on these cases, so this shows what the logic reaches, not unseen accuracy.

Keyword extractor, no LLM, so anyone can reproduce it. 7

CHECKED BY PEOPLE, NOT ONLY BY SCRIPTS

Every journey tested, every answer read

5 / 5

Sample journeys pass

Message, clicked link, call, breach, small-business incident

20 / 20

Safe in manual review

First pass 18/20; we fixed the 2 gaps and re-scored

4 / 6

Understood by outsiders

One listener went to the police first, not the bank

A failure we found and fixed

“Our supplier changed bank account, can I pay?” got the “already paid” answer first. Now “can I pay?” means not paid yet, and the reply starts with “Don’t pay. Verify by phone.”

555 automated tests, 335 test-set predictions validated. 8

INSIDE DNA'S SERVICES

Built to plug into the DNA app

Part	Prototype today	In DNA's services
Entry point	Web app	"Is this a scam?" in the DNA app and Oma portal
Identity	Mock login	Existing DNA login, no sign-up
Signals	Simulated, behind consent	SIM/eSIM events, logins, call warnings
Handoff	Simulated summary	Ticket to DNA customer or business support
Link check	Server-side fetch	Isolated scanner plus DNA threat feeds

One API, **/api/sessions**. Each simulated adapter swaps for a real one without touching the rules.

PACKAGING, A HYPOTHESIS

Safety help free for everyone. Extras in the bundle.

FREE, IN THE DNA APP

€0

Next step, link check, Rescue Mode, human handoff

DNA SECURITY BUNDLE

€2.99 /mo

DNA signal checks, decision report, priority help

DNA BUSINESS SECURITY

€9 /mo

Per company: staff accounts, invoice-fraud checks

For DNA: fewer scam support calls, a reason to keep the security bundle, and a head start as the EU proposes fraud-prevention duties for telecom operators.

Prices are a hypothesis to test, not a result. 10

What should I do now? SafeStep answers.

Next: native-speaker review of Finnish and Swedish, a clearer “bank first” order, and a pilot inside the DNA app.

Live demo
safestep.mehrabdev.com

Video
youtu.be/uNQ2D_KD0XE

Code
github.com/Arsham1ho/TBL_Hacketon_2026